

MULTIMODAL RECOGNITION

Week 2: Object Segmentation

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MASTER IN
COMPUTER VISION
Barcelona



GitHub Repository

https://github.com/JordiVentosa/C5_Team3

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Dataset description and Experimental Setup

Volume: 21 video sequences containing a total of 8,008 frames.

Data Used: Exclusively the official training dataset (only split with Ground Truth available).

Splitting Strategy:

To prevent data leakage and accurately measure model behavior, we split the data by video sequence, following the configuration from the original KITTI-MOTS paper.

Training Split – 12 Sequences (57%)

00 01 03 04 05 09 11 12 15 17 19 20

Validation Split – 9 Sequences (43%)

02 06 07 08 10 13 14 16 18

Classes & Distribution:

- Class 1 (Cars): 26,860 detections.
- Class 2 (Pedestrians): 11,415 detections.
- Class 10 (Ignore Region): **Discarded**.

Note: Cars outnumber pedestrians by more than 2:1. This imbalance should be considered when analyzing the model's behavior.



SAM Version Used: SAM-ViT-Huge (636M parameters) from Hugging Face.

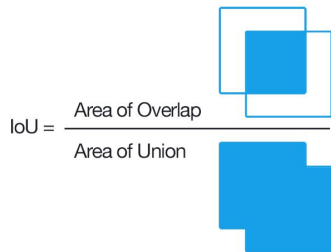
Metric description

To evaluate our models, we used the official COCO evaluation framework via *pycocotools* instead of the default KITTI-MOTS metrics. This provides a standardized benchmark across multiple scales and IoU thresholds.

Mathematical Foundation:

IoU (Intersection over Union):

Determines detection validity by measuring the overlap between the predicted bounding box and the Ground Truth.



COCO Evaluation:

AP: Average Precision across 10 IoU thresholds (0.50 to 0.95, step 0.05).

AP₅₀ & AP₇₅: Evaluated at a single IoU threshold.

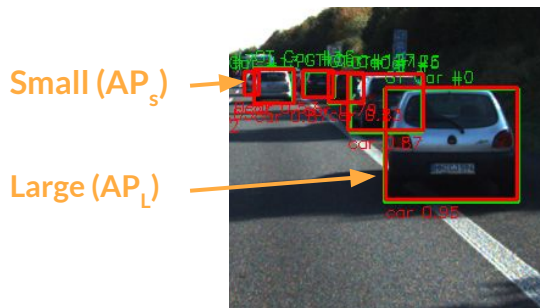
AR: Average Recall across 10 IoU thresholds (0.50 to 0.95, step 0.05).

Scale-Specific Metrics:

AP/AR_s: Small objects (area < 32² pixels)

AP/AR_M: Medium objects (32² < area < 96²)

AP/AR_L: Large objects (area > 96²)



A. SAM inference with point prompts

A. SAM inference with point prompts

Quantitative

Prompt generation from ground-truth masks: For each KITTI-MOTS instance, the code builds interactive SAM prompts directly from the object mask GT by selecting points inside the instance or around its boundary.

Four prompt strategies are compared: (1) one positive point at the object center, (2) one random positive point inside the mask, (3) multiple positive points to better cover elongated or partially occluded objects, and (4) one positive point plus one negative point outside the object near the border.

Analysis: Using 3 random positive points gives by far the best results across all metrics. Might be that multiple points improve object coverage and lead to much better results.

Type of Prompt	Center Point	Random Point	3 Random Points	Pos Point + Neg Point
AP	0.119	0.093	0.471	0.123
AP (small)	0.159	0.123	0.345	0.164
AP (medium)	0.093	0.090	0.569	0.098
AP (large)	0.220	0.114	0.612	0.182
AR	0.258	0.234	0.607	0.263
AR (small)	0.381	0.344	0.473	0.385
AR (medium)	0.149	0.161	0.686	0.164
AR (large)	0.305	0.168	0.724	0.272

A. SAM inference with point prompts

Qualitative 1



In this simple scene, all prompt variants are able to segment the pedestrian, but multiple positive points produce a more complete and stable mask.

Single-point prompts work, although they are more sensitive to the exact point location, especially around thin structures such as legs.

Prompt: 1 positive + 1 negative



Prompt: 3 positive points



Prompt: 1 positive centered point



Prompt: 1 random positive point



A. SAM inference with point prompts

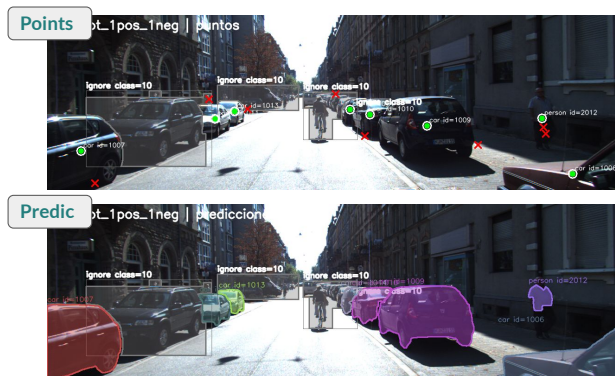
Qualitative 2



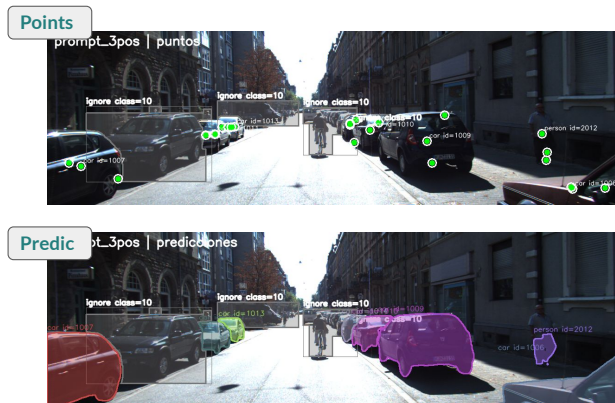
In this other scene, the difference between prompting strategies becomes not that clearer. In this case, 3 positive points recover less than single-point in the person instance.

By contrast, the car segmentations are perfectly segmented by all the prompt methods.

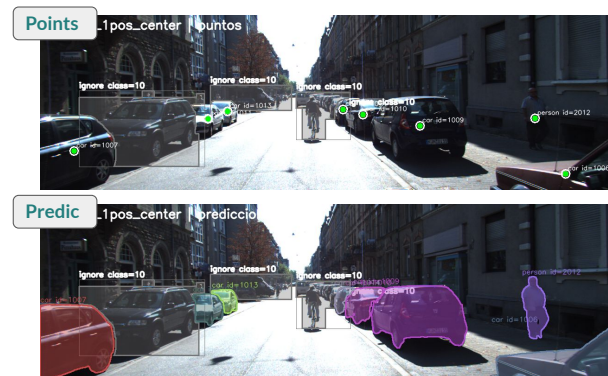
Prompt: 1 positive + 1 negative



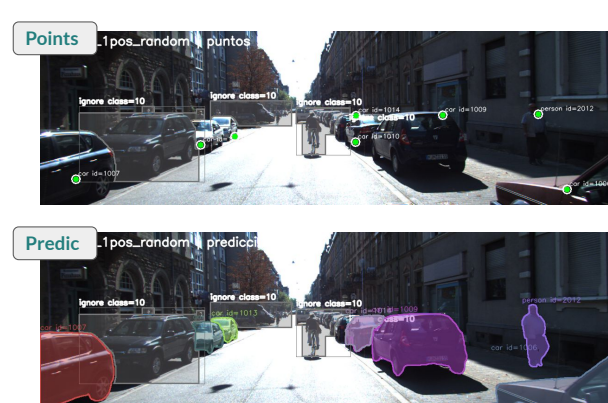
Prompt: 3 positive points



Prompt: 1 positive centered point



Prompt: 1 random positive point



B. Grounded SAM inference with text prompts

B. Grounded SAM inference with text prompts

Grounding DINO is used with a box threshold of 0.3 and a text threshold of 0.25

Quantitative

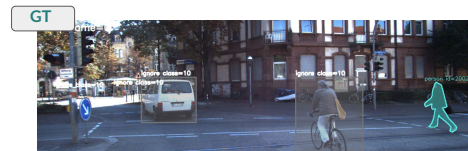
- Grounded SAM is not a single model. It is a two-stage pipeline that combines Grounding DINO and SAM.
(text prompt → Grounding DINO finds boxes → SAM turns boxes into masks)
- The code evaluates two prompt settings for Grounding DINO: one runs separate detections with "**car.**" and "**person.**", and the other runs separate detections with "**car.**" and "**pedestrian.**".
In both settings, the detections are merged, each box is passed to SAM for segmentation, and the combined results are overlaid and evaluated together against the ground truth.
- Grounding DINO prompts usually work best with lowercase letters and a dot mark.
- Overall, Grounded SAM performs average. The low AP_{small} and high $AP_{\text{mid/large}}$ demonstrates that it detects better the big in-front objects, but fails a lot with distant cars and persons.

Model	AP	AP (small)	AP (mid)	AP (large)	AR	AR (small)	AR (medium)	AR (large)
Gro. SAM	0.450	0.221	0.571	0.514	0.523	0.372	0.631	0.698

B. Grounded SAM inference with text prompts

Qualitative 1

- Grounding DINO usually works best with lowercase letters and a dot mark.

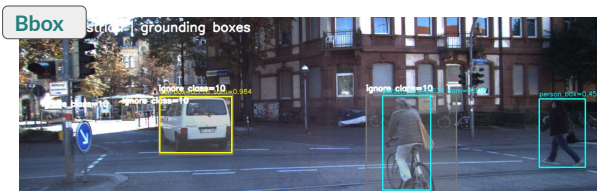


Prompt: “car.” + Prompt: “person.”



For this relatively simple example, both text prompts (“person.” and “pedestrian.”) produce very similar detections and masks.

Prompt: “car.” + Prompt: “pedestrian.”



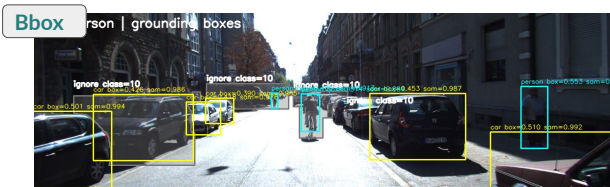
This suggests that the pipeline is robust to small wording changes when the objects are clearly visible and well separated.

B. Grounded SAM inference with text prompts

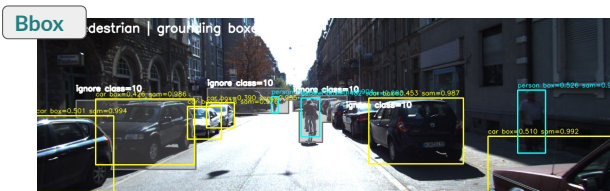
Qualitative 2

- Grounding DINO usually works best with lowercase letters and a dot mark.

Prompt: “car.” + Prompt: “person.”



Prompt: “car.” + Prompt: “pedestrian.”



In the more complex scene, Grounded SAM still separates cars and pedestrians reasonably well with both text prompts.

Most errors appear on small, distant, or partially occluded objects, while the prompt wording has only a minor qualitative effect.

C. SAM inference with YOLO bboxes

C. SAM inference with YOLO bboxes

Quantitative

From previous week we got that our best model was YOLO26x without fine tuning, so our best model and the pre-trained model without fine tuning are the same. However, to see how the quality of bounding boxes affect the final metrics and predictions we have decided compare this results to a smaller version of this same model: YOLO26n, which has less accuracy in exchange for a huge reduction of parameters.

As shown in the table, the smaller model has both lower precision and lower recall compared to the larger model. The main difference between the two lies in their ability to detect small and medium-sized objects, where the smaller model struggles the most. However, their performance on large objects is similar.

This suggests that bounding box quality strongly affects recall and precision for small and medium objects, since small localization errors can easily miss part of the object or include background. In contrast, large objects are less affected because these errors represent a much smaller portion of their area.

Model	AP	AP (small)	AP (medium)	AP (large)	AR	AR (small)	AR (medium)	AR (large)	Num. Parameters
YOLO26x	0.503	0.249	0.641	0.768	0.592	0.360	0.721	0.889	59M
YOLO26n	0.294	0.059	0.404	0.751	0.344	0.083	0.450	0.857	2M

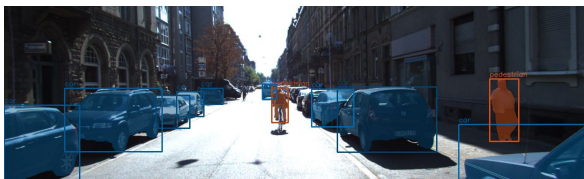
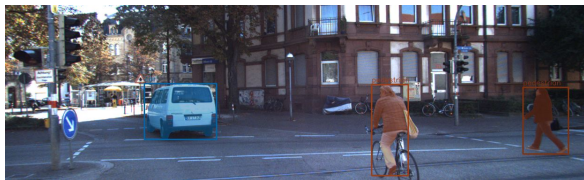
C. SAM inference with YOLO bboxes

Qualitative

GT

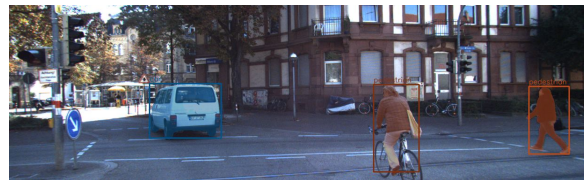


YOLO26x



Gives accurate bounding boxes and is able to detect even smaller or occluded objects, allowing for better segmentation. However, it detects more objects than given on the ground truth, increasing the number of false positive detections.

YOLO26n



Bounding boxes are less accurate, sometimes entering into the object (cyclist from first image), hurting SAM performance. It struggles with occluded objects such as the left white car, which is not detected.

D. Compare Grounded SAM with YOLO bboxes

D. Compare Grounded SAM with YOLO bboxes

Quantitative

Both approaches rely on SAM for mask generation, but differ in the object detection stage. While the first pipeline uses YOLO26x as the detector, Grounded SAM relies on Grounding DINO to generate bounding boxes from text prompts.

As we can clearly see on the table, using SAM with YOLO26x outperforms Grounded SAM with a 0.053 gap on AP and a 0.069 gap on AR. Unlike the drop observed when comparing YOLO26x and YOLO26n on task c, which primarily affected small objects, the reduction here occurs across all object sizes. While small-object AP decreases slightly (0.249 -> 0.221), medium and large objects experience larger drops (0.641 -> 0.571 for medium, 0.768 -> 0.514 for large), indicating a general reduction in detection quality rather than a scale-specific issue.

These results are expected since YOLO26x is trained on this specific set of objects, whereas Grounding DINO is an open-vocabulary detector designed for the detection of arbitrary objects from prompts, providing it flexibility but having lower performance than specialized models for a closed set of objects.

Model	AP	AP (small)	AP (medium)	AP (large)	AR	AR (small)	AR (medium)	AR (large)
YOLO26x	0.503	0.249	0.641	0.768	0.592	0.360	0.721	0.889
Gro. SAM	0.450	0.221	0.571	0.514	0.523	0.372	0.631	0.698

D. Compare Grounded SAM with YOLO bboxes

Qualitative 1

As we can see in the images on the right, Grounded SAM appears to struggle with detecting people and is not able to segment any of the pedestrians crossing the street. On the other hand, the model using YOLO26x is able to detect both the pedestrians crossing the street and the person on the bicycle on the right. Even though this cyclist is not labeled in the ground truth, the detection suggests that the YOLO26x model is more reliable when detecting people.

We can also observe that both models are able to detect the car in the front and the one coming from the opposite lane. However, both models struggle to detect heavily occluded objects, such as the car in the top-right corner.

GT



YOLO26x



Gro. SAM



D. Compare Grounded SAM with YOLO bboxes

Qualitative 2

To ensure that the previous results were not simply due to the small size of the pedestrians, we can also observe how similar results are obtained when they are much closer to the camera. As shown, the model using YOLO26x is able to detect every labeled pedestrian, even when they are close together and slightly overlapping, as with the two pedestrians in the middle. On the other hand, Grounded SAM does not seem to be able to segment pedestrians when they overlap.

As before, both models are able to correctly segment the cars appearing in the image, showing that both models are robust on their detection.

GT



YOLO26x



Gro. SAM



E. Fine-tune the Prompt-Decoder of SAM

E. Fine-tune the Prompt-Decoder of SAM

Strategy

We fine-tuned the decoder of the SAM Base model using bounding box prompts, which were computed from the ground truth masks using min/max coordinates. We chose the Base version instead of the Huge one due to computational constraints, allowing us to perform more extensive experiments.

Data augmentation was organized into different profiles (as shown in the table) to evaluate various strategies, such as photometric transformations.

First, we performed a full fine-tuning. Then, using the best-performing profiles from those experiments, we applied LoRA.

Base Weights: SAM ViT Base

Loss: Same as SAM (Pondered Focal + Dice)

LR: 1e-5 for FT | 1e-4 for LoRA

Batch Size: 8

Training Profiles for Data Augmentation		
Profile	Strategy	Transformations Applied
P0	No DA	No augmentations; standard training
P1	Photometric	ColorJitter, Gaussian Noise, Brightness/Contrast
P2	Geometric	Horizontal Flip, Random Resized Crop
P3	Combined	P1 + P2 + ShiftScaleRotate and CoarseDropout

Training Approaches

Full Fine-Tuning (FT): Applied to all profiles (P0-P3), freezing all the model except the decoder.

LoRA: Selectively applied only to P0 and P1 based on their superior performance in initial trials.

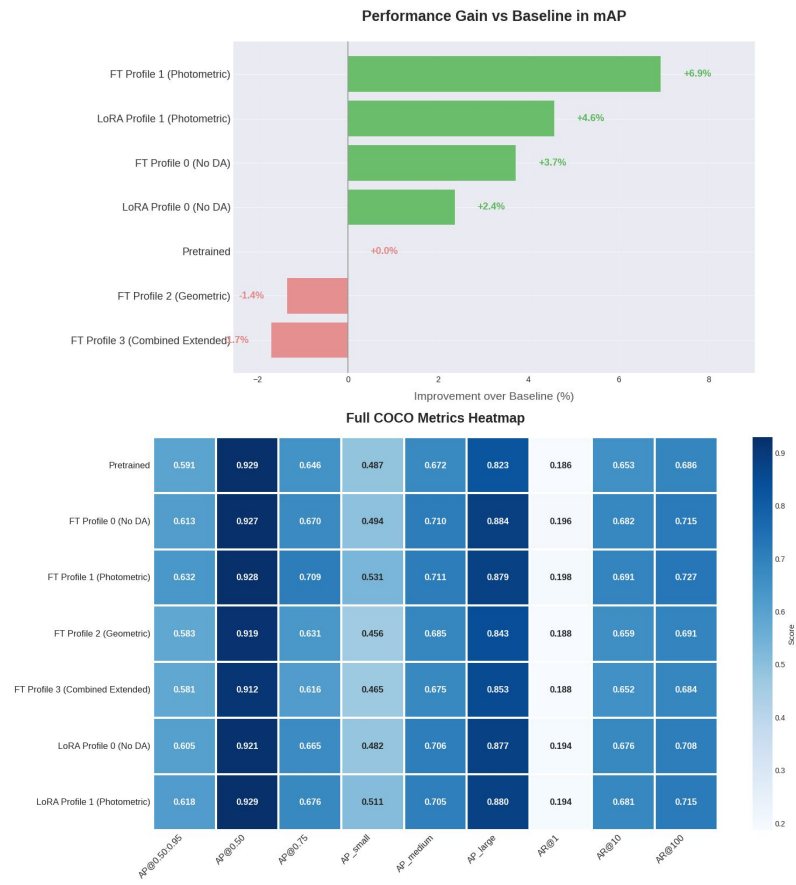
E. Fine-tune the Prompt-Decoder of SAM

Quantitative Results

Photometric augmentation proved to be the most effective configuration, whereas geometric and combined profiles negatively impacted performance. This suggests the model is highly sensitive to geometric distortions. This is expected since photometric changes occur naturally, whereas a rotated car is rarely encountered in real-world driving scenarios.

Although LoRA underperformed compared to full Fine-Tuning, the results are strong enough to justify its use for resource-constrained environments.

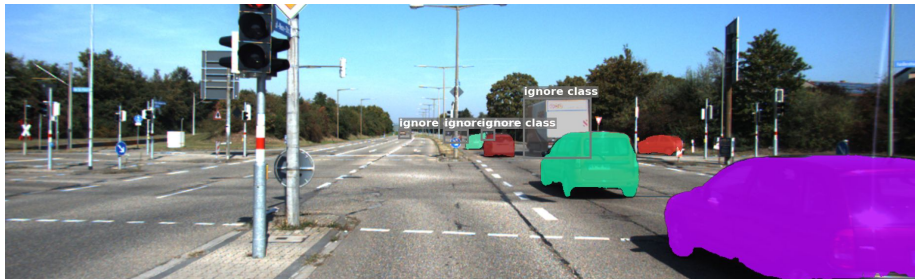
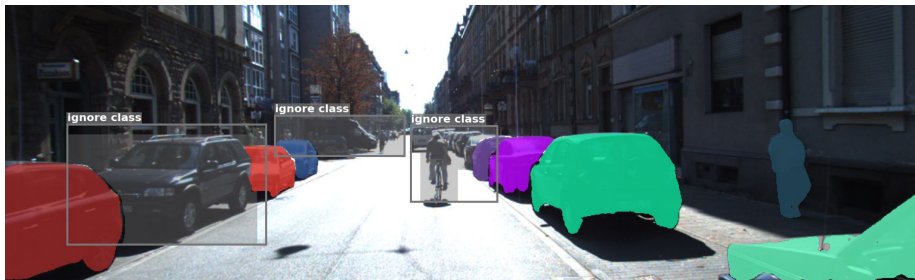
mAP SAM Base	Standard Finetuning				LoRA	
	Profile 0	Profile 1	Profile 2	Profile 3	Profile 0	Profile 1
0.591	0.613	0.632	0.583	0.581	0.605	0.618



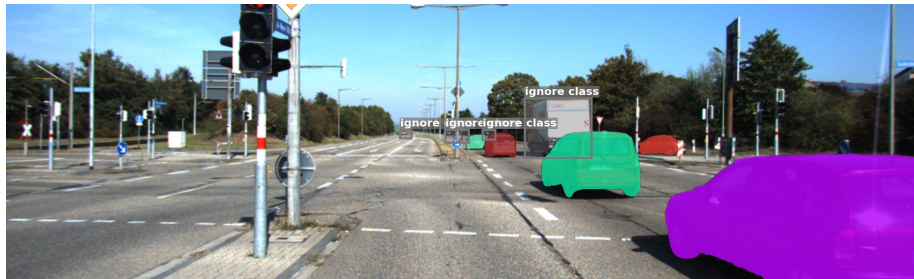
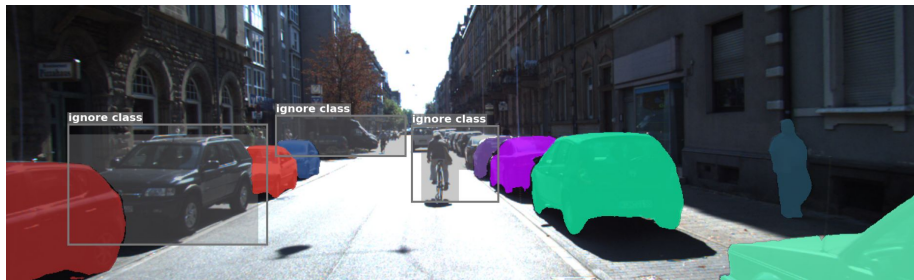
E. Fine-tune the Prompt-Decoder of SAM

Qualitative Examples

SAM Base



Fine-tuned Profile 1



In general, both models perform similarly, but the fine-tuned version produces smoother and better-adjusted masks.

E. Fine-tune the Prompt-Decoder of SAM

Comparison: Fine-Tuned vs. Baseline across different prompts

Since we used SAM Base for fine-tuning instead of the Huge version, we re-ran the previous tests on both the baseline SAM Base and our Fine-Tuned model (using Profile 1).

The prompts tested are the following:

- BBox derived from the GT mask's min/max coordinates (Perfect BBox)
- BBox predicted by YOLO26x
- Center point of the GT mask
- Single random point inside the GT mask
- Three random points inside the GT mask
- One positive and one negative point based on the GT mask

Our experiments show consistent improvements for the Fine-Tuned model across almost all prompt types. The only exception is the *3 random points* prompt, where performance slightly decays compared to the Base model. As expected, the improvement with YOLO BBoxes is smaller than with Perfect BBoxes, since imperfect detections introduce misalignment. Because the *3 random points* prompt is the most informative point-based strategy, it suggests that for highly specific point-prompting in KITTI-MOTS, the fine-tuned version may not completely outperform the robust baseline.

Models in different prompts	mAP			AR		
	SAM Base	SAM FT P1	Relative Improve	SAM Base	SAM FT P1	Relative Improve
Perfect BBox	0.591	0.633	+7.20%	0.686	0.730	+6.41%
YOLO BBox	0.322	0.331	+2.79%	0.469	0.484	+3.20%
Center Point	0.096	0.125	+29.97%	0.216	0.250	+15.74%
Random Point	0.074	0.096	+30.61%	0.190	0.216	+13.68%
3 Random Points	0.407	0.385	-5.50%	0.554	0.543	-1.99%
Pos Point + Neg Point	0.101	0.130	+28.84%	0.225	0.252	+12.00%

F. Pre-trained SAM and the fine-tuned version domain shift

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Dataset, Experiment and Quantitative Results

Dataset: We used a fraction of the iSAID dataset, which contains satellite images. We chose this dataset because it is extremely challenging for SAM. It consists of aerial images (which are infrequent in SAM's original training data) and contains very small objects (which SAM typically struggles with). Every experiment was performed using SAM Base and its fine-tuned versions.

Hypothesis: The high difficulty of this dataset should allow us to clearly observe the performance differences and better study the domain shift. We evaluated the baseline SAM, Full Fine-Tuning (Profile 1), and LoRA (Profile 1). Our initial assumption was that LoRA would perform better than full FT due to its resistance to catastrophic forgetting, though neither would outperform the pre-trained baseline.

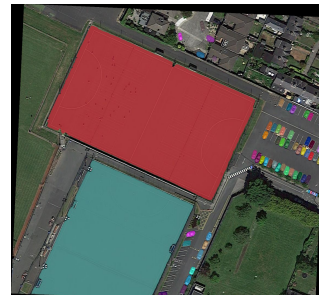
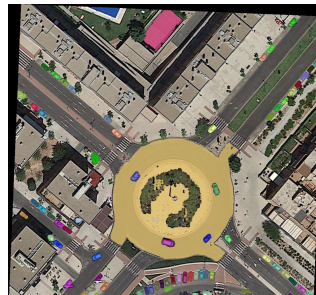
Model	AP	AP (small)	AP (medium)	AP (large)	AR	AR (small)	AR (medium)	AR (large)
SAM Base	0.287	0.226	0.442	0.675	0.456	0.383	0.601	0.810
FT Profile 1	0.260	0.213	0.376	0.620	0.427	0.370	0.528	0.780
LoRA Profile 1	0.247	0.200	0.363	0.616	0.417	0.356	0.521	0.794

Results: The evaluation shows that SAM Base outperforms both fine-tuned versions on iSAID, as expected. Contrary to our hypothesis that LoRA would better preserve pre-trained knowledge, it suffered the largest performance drop. This suggests that LoRA over-specialized in the basic shapes and perspective of a car-mounted camera (KITTI-MOTS). When applied to an aerial view, these LoRA adapters acted as a rigid filter, hindering performance.

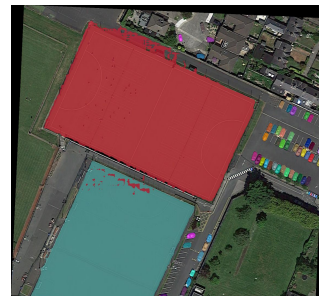
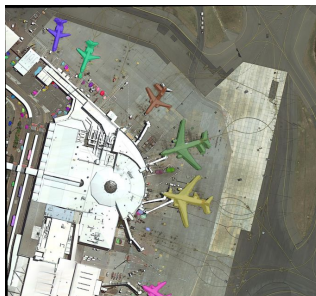
F. Pre-trained SAM and the fine-tuned version domain shift

Qualitative

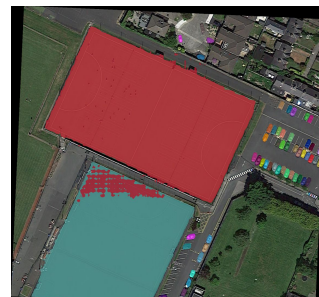
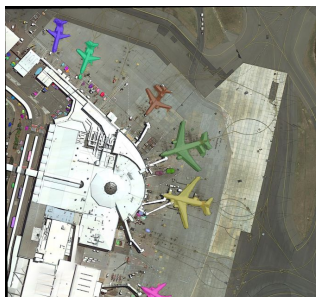
SAM Base



Full
Fine-tuned



LoRA



G. Analysing the differences between the prompts

G. Analysing the differences between the prompts

Quantitative

We will be using the Huge version of SAM to do the comparisons, since it gives the best results.

As we can observe on the table to the left, the type of prompt strongly influences performance across different sizes:

- **Bounding box prompts:** they achieve the highest overall AP (0.503) and excel at medium and large objects, providing the best recall and precision for these sizes.
- **Points prompts:** they perform best for small objects, with higher AP (0.345) and AR (0.473) than bounding boxes or text prompts, indicating that point guidance is particularly effective for tiny objects.
- **Text prompts:** they perform the weakest overall, struggling especially with small and large objects, making it a flexible but not too performant option.

Metric vs. Prompt	3 Random Points (Point prompt)	Grounded SAM (text prompt)	YOLO26x BBox (Bbox prompt)
AP	0.471	0.450	0.503
AP (small)	0.345	0.221	0.249
AP (medium)	0.569	0.571	0.641
AP (large)	0.612	0.514	0.768
AR	0.607	0.523	0.592
AR (small)	0.473	0.372	0.360
AR (medium)	0.686	0.631	0.721
AR (large)	0.724	0.698	0.889

Among all these prompt options, the most robust are the bounding box prompts. As a future improvement, a hybrid approach that combines both point and bounding box prompts could enhance performance: point prompts would help with small objects, while bounding box prompts would maintain performance for large objects.

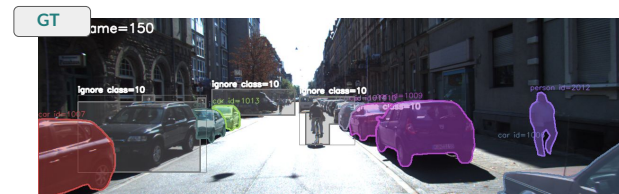
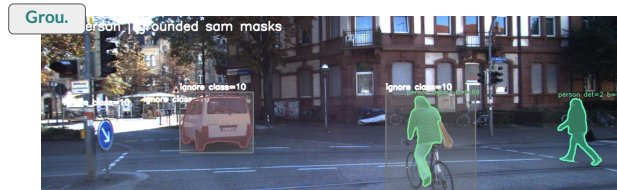
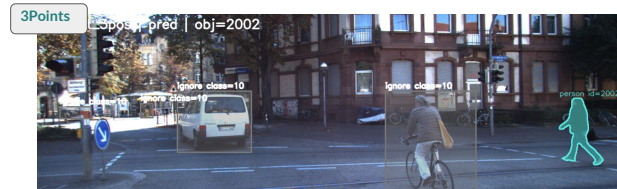
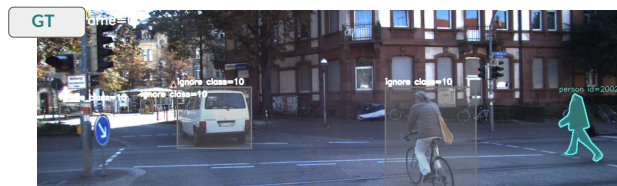
G. Analysing the differences between the prompts

Qualitative

Point prompts produce masks that are often broken when it comes to people, since the minimal information provided by them do not allow SAM to consider the whole pedestrian, but just one part of it.

Text prompts struggle to detect some overlapping objects, such as one of the labeled parked cars on the right.

Bbox prompts seem more reliable in general, being able to detect and segment objects that are slightly overlapping robustly. However they struggle with smaller objects, such as the cyclist far away on the left side of the lane, which even though it is labeled as to ignore, is still a pedestrian.



H. Semantic Segmentation

H. Semantic Segmentation

Quantitative

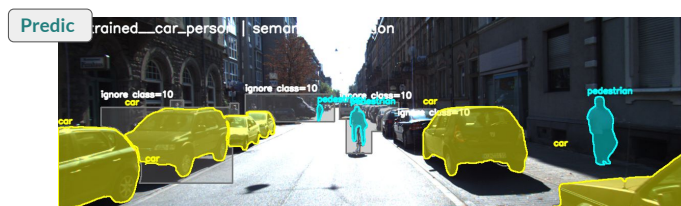
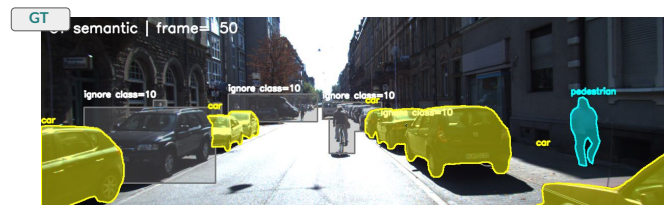
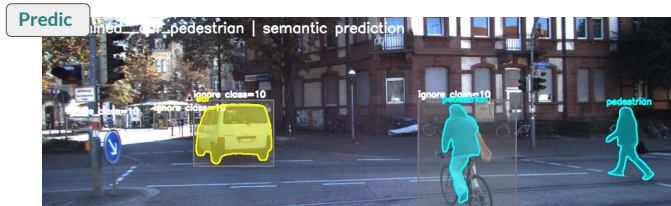
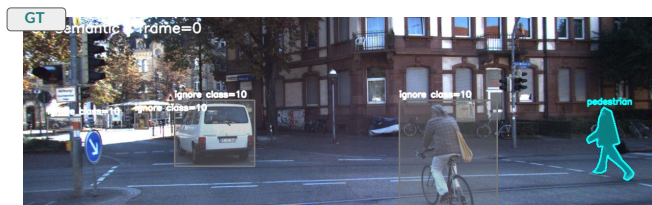
- **GT preparation:** Converts instance masks into a semantic ground truth by class (car, pedestrian), and separates the ignore region.
- **Prediction and evaluation:** Uses Grounded SAM (Grounding DINO + SAM) with text prompts to generate semantic segmentation, merges masks by class into a final semantic map, and computes per-class and global metrics.
- **Analysis:** Performance improves significantly with object size and is consistently higher for Person than for Car, while small objects remain the main challenge

Class	AP	AP (small)	AP (medium)	AP (large)	AR	AR (small)	AR (medium)	AR (large)
Global	0.633	0.422	0.739	0.874	0.691	0.523	0.786	0.909
Person	0.717	0.497	0.807	0.922	0.761	0.587	0.836	0.949
Car	0.549	0.348	0.671	0.827	0.621	0.459	0.735	0.868

H. Semantic Segmentation

Qualitative

The predicted semantic maps are globally consistent with the ground truth, especially for large cars and clearly visible pedestrians. The main limitations appear on small or distant instances and object boundaries, where some regions are slightly imprecise or missed.



Summary Slide



Quantitative :

Model	AP
YOLO26x + SAM	0.503
GD SAM	0.450
3 Random Points	0.471



Conclusions:

- **Bounding box prompts** outperform other prompt types but perform slightly worse than **point prompts** for small objects.
- Specialized detectors like YOLO26x outperform open-vocabulary ones like Grounding DINO across all object sizes and metrics for closed sets of objects.
- Bounding box quality impacts severely segmentation performance in SAM.



Qualitative:



Fine-tuning and Domain Shift (iSAID):

- **Best FT Strategy:** Full Fine-Tuning with Photometric Data Augmentation.
- **LoRA Limitations:** Underperformed Full FT and failed to prevent catastrophic forgetting.
- **Domain Shift (iSAID):** Models over-specialized to KITTI-MOTS; Baseline SAM remained the most robust out-of-domain.